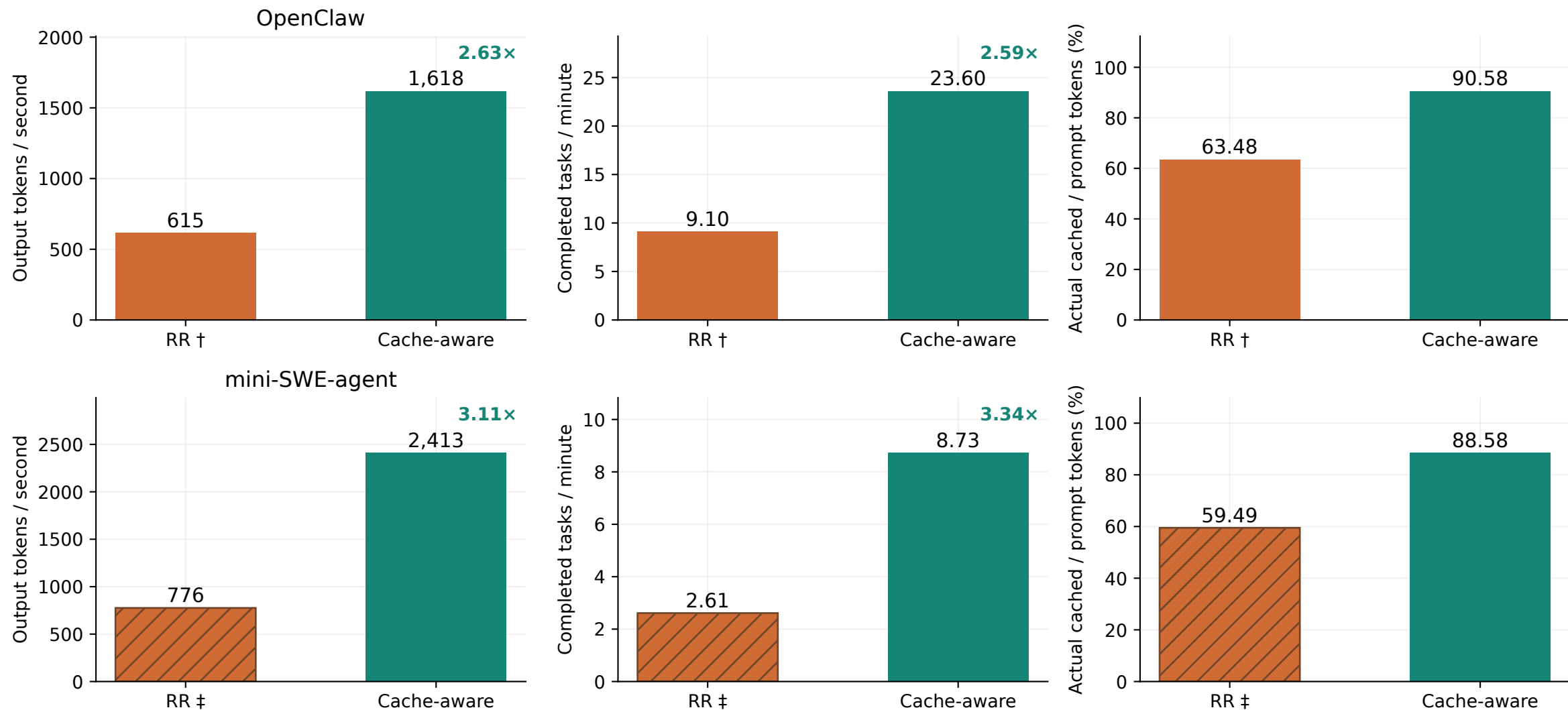
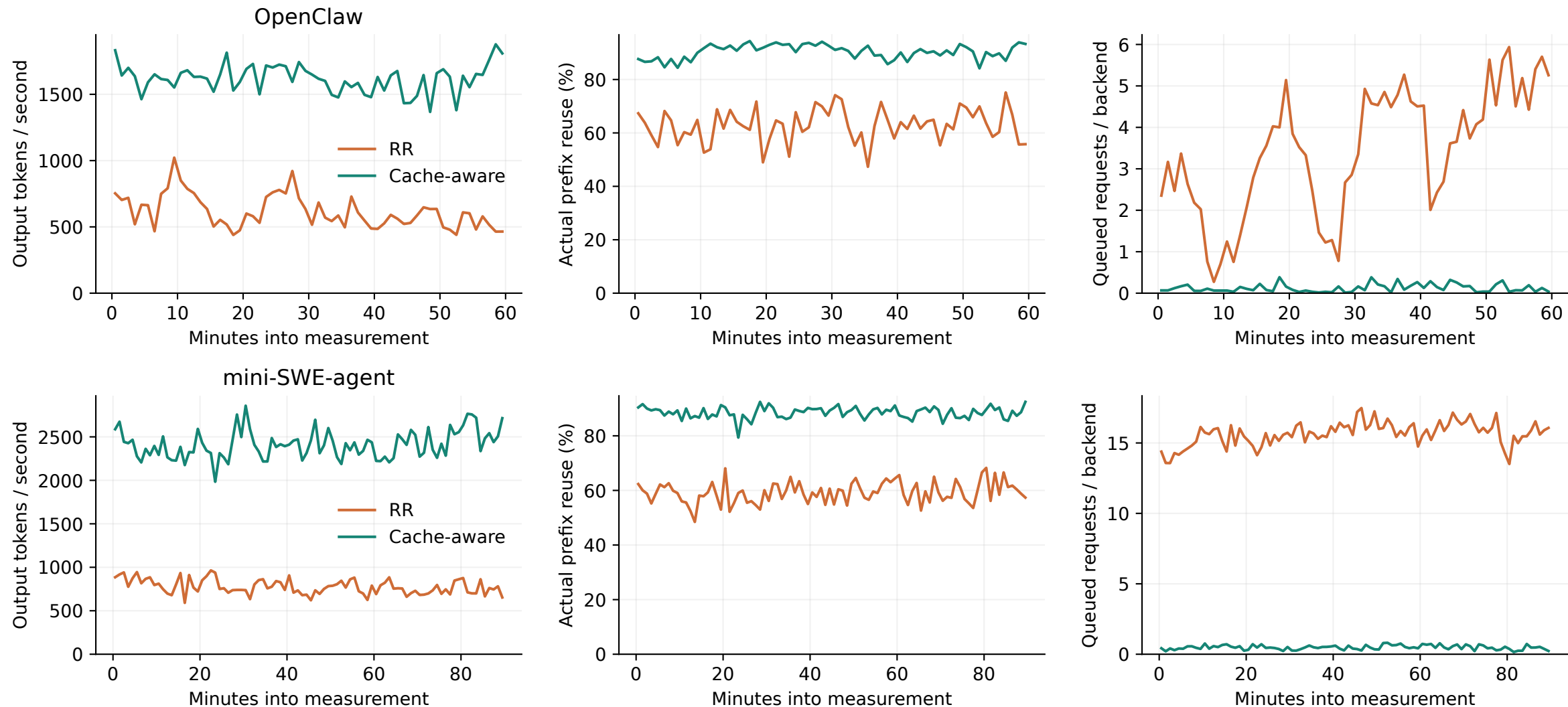


Cache-aware routing improves throughput at the same 1+4 placement



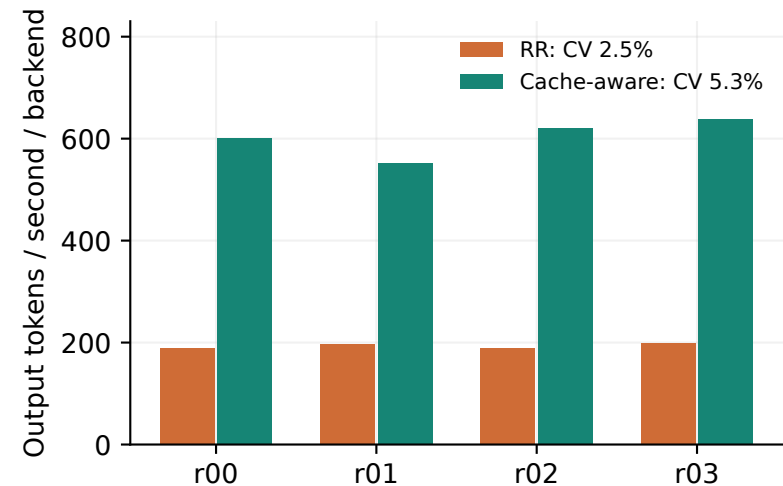
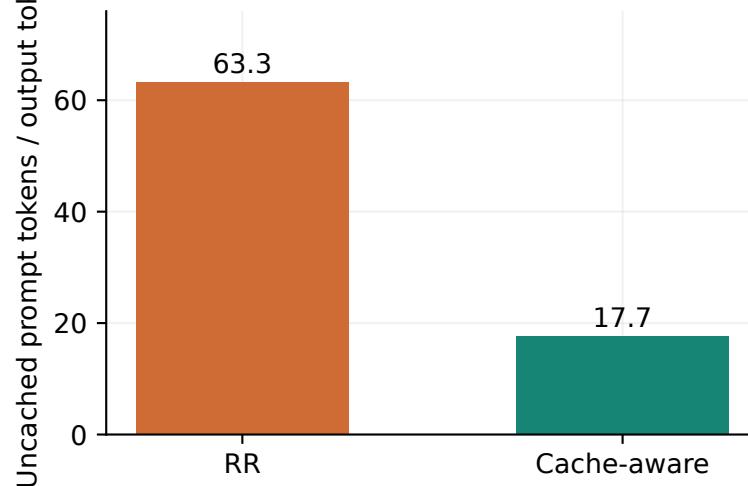
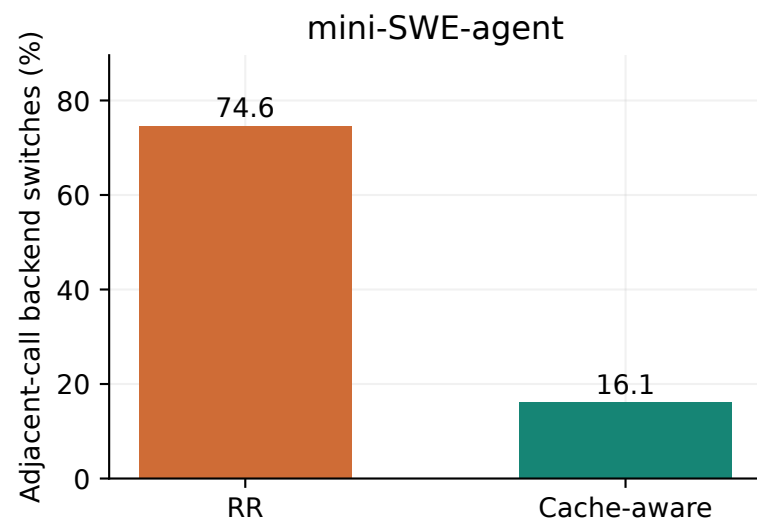
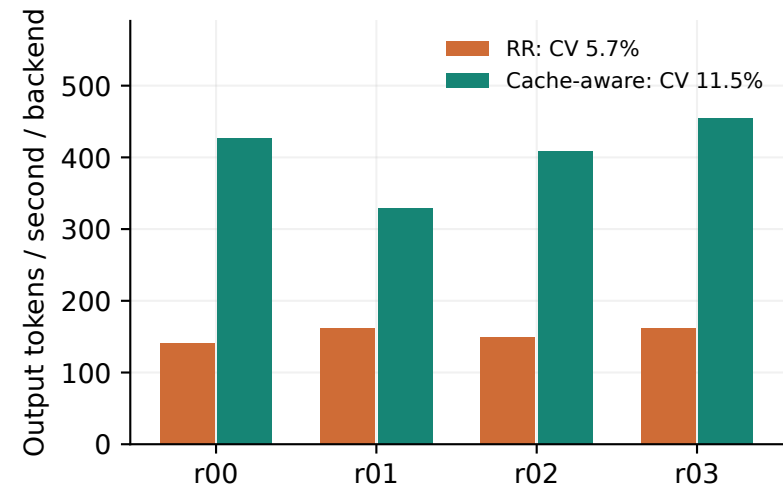
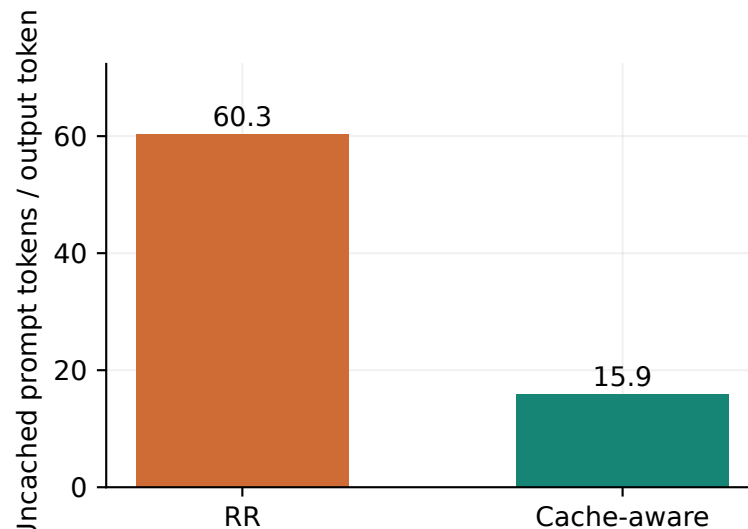
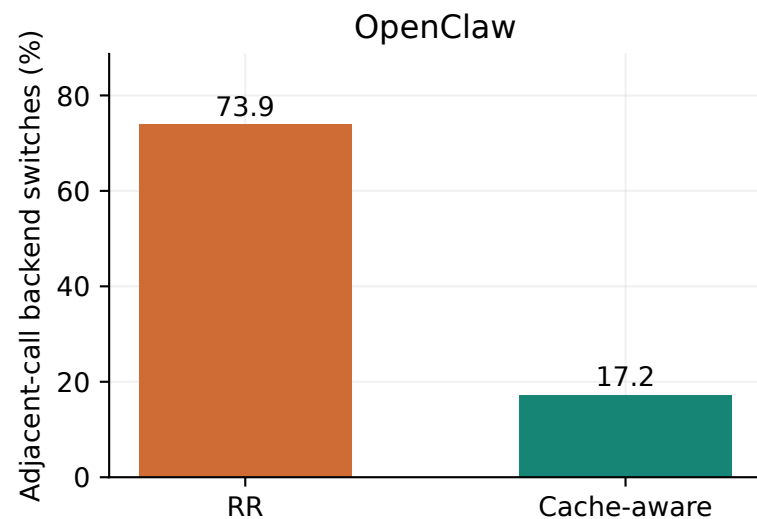
Full windows: OpenClaw 60 min, mini-SWE 90 min. † RR is nonsteady. ‡ Full-window counts recovered; drain timed out.
Single run per point; error bars are unavailable. Mini-SWE RR task/LLM latency cohorts are censored and omitted.

Routing trajectories: reuse, throughput, and backend queues



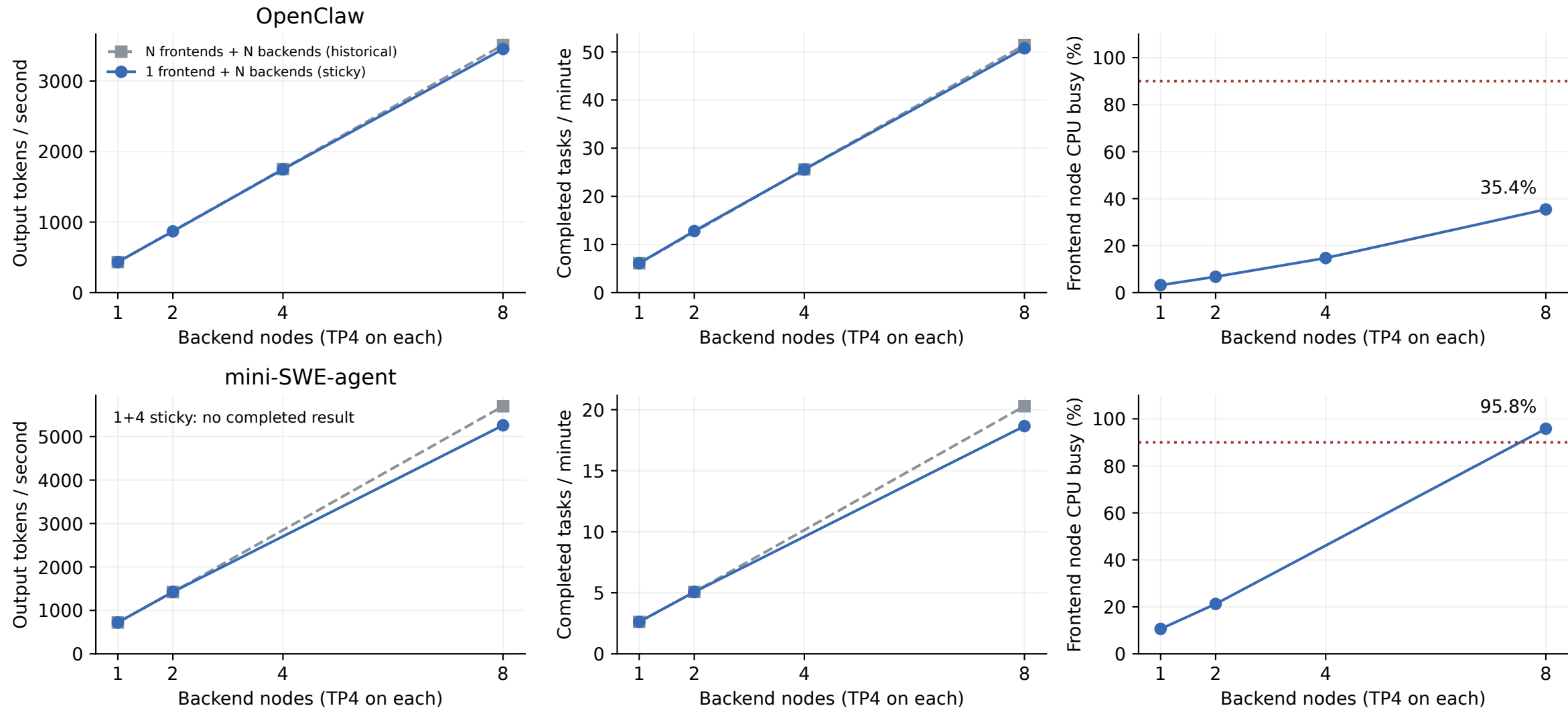
60-second bins; queued requests use all backend samples. Curves cover measurement only, excluding warmup/drain.
RR nonstationarity and mini-SWE drain cancellation limit steady-state capacity / tail-latency claims.

Routing changes locality and the amount of uncached prefill work



Switch rate: adjacent observed completed calls of tasks with ≥ 8 window calls; weighted by adjacent-call pairs.
Uncached input/output is a window-level work ratio, not a matched-request estimate. Cache-aware is not strict task stickiness.

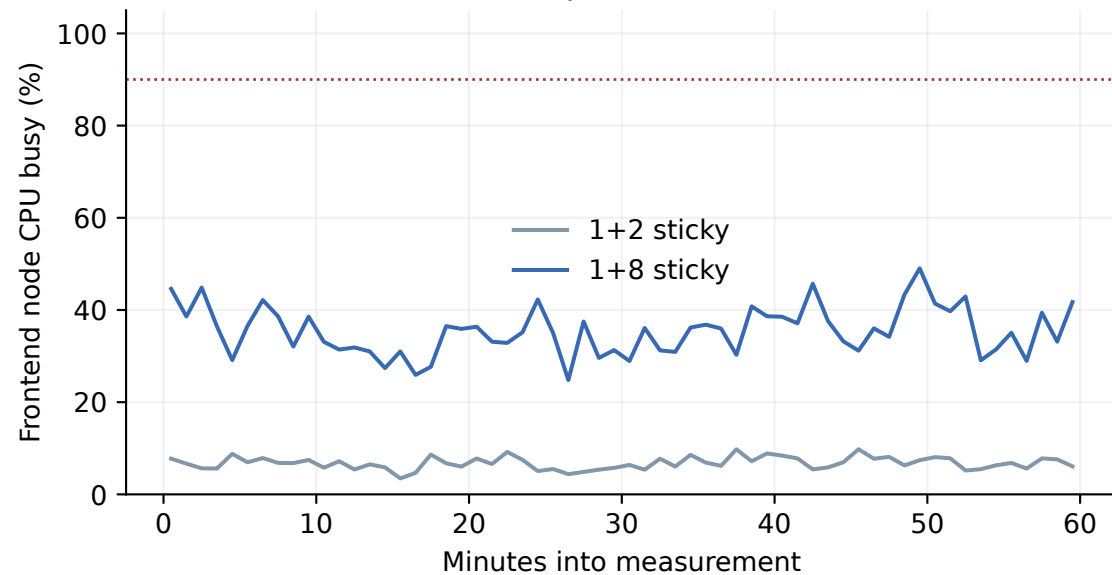
One frontend scales differently for the two workloads



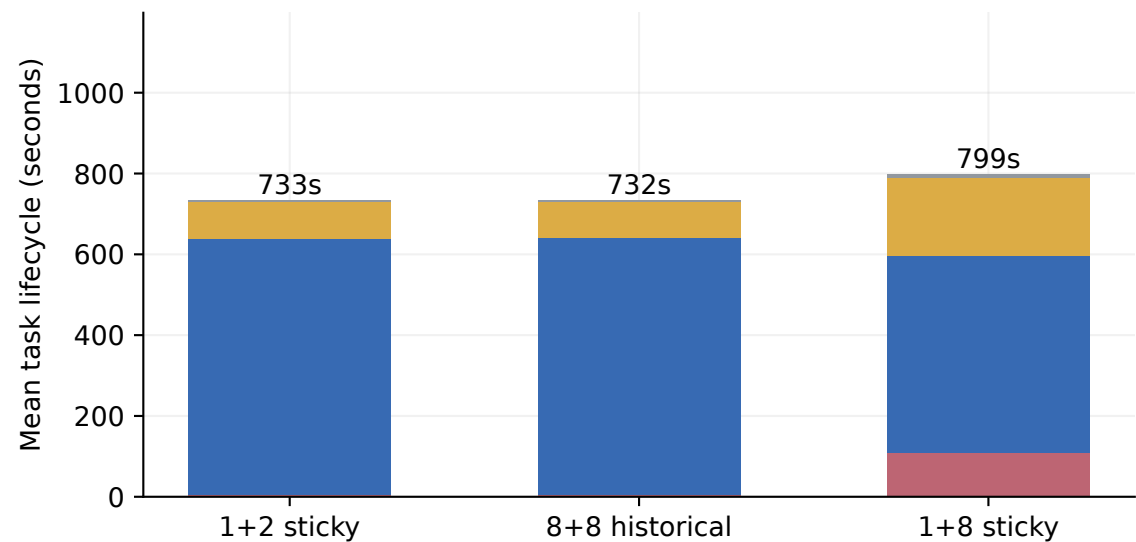
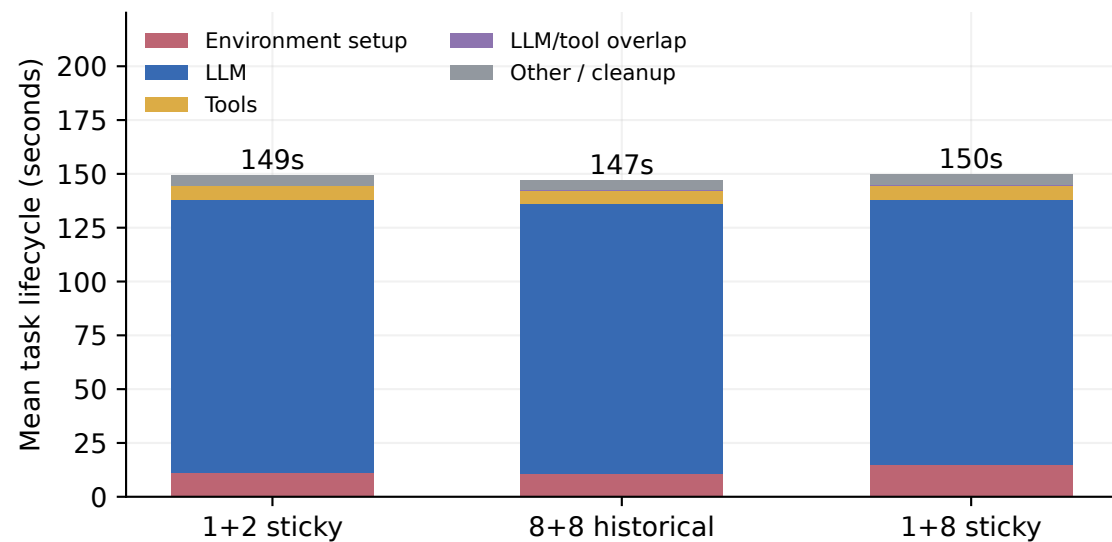
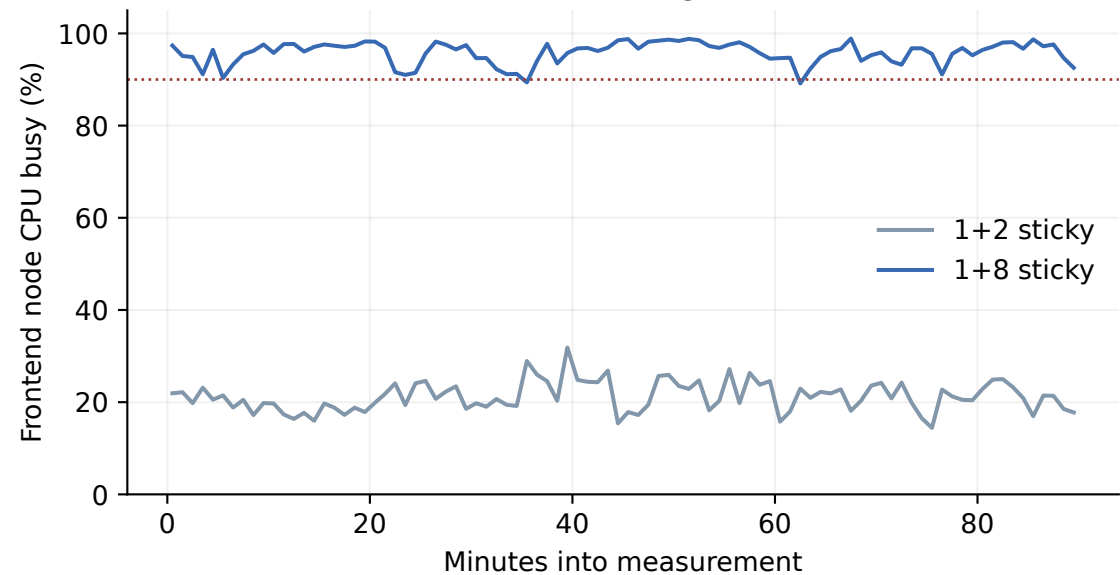
All markers are measured; segments merely connect available points. N=1 is the shared historical baseline.
Historical balanced runs differ in HTTP/build policy: useful capacity references, not controlled frontend-count comparisons.

Frontend pressure and where task time is spent

OpenClaw

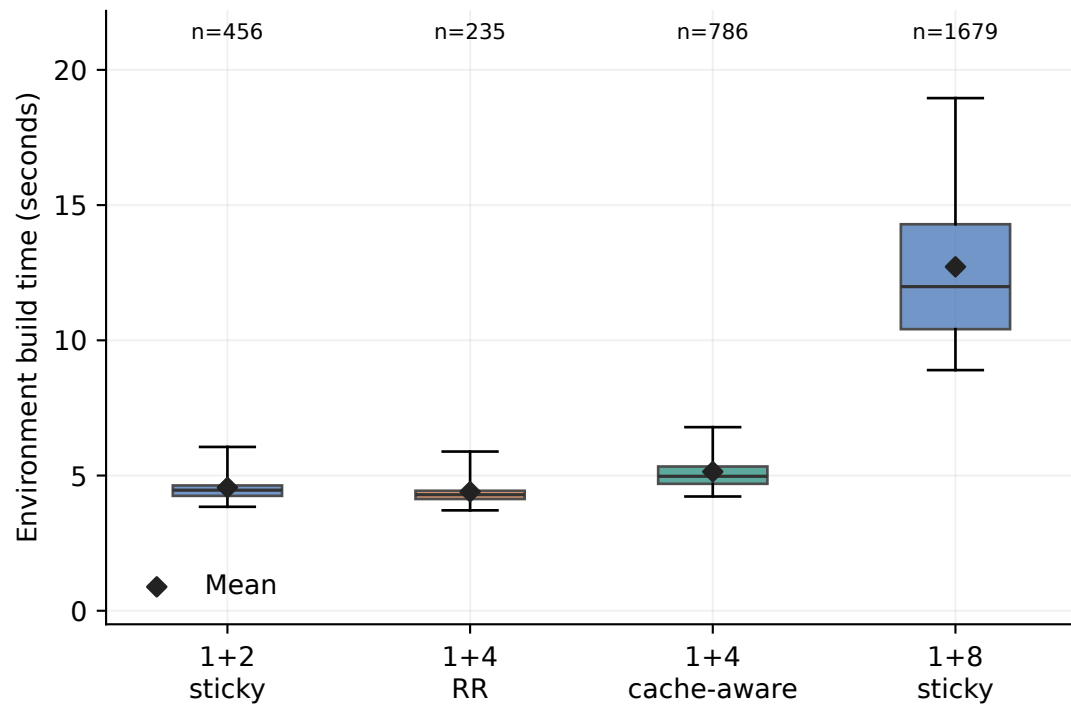
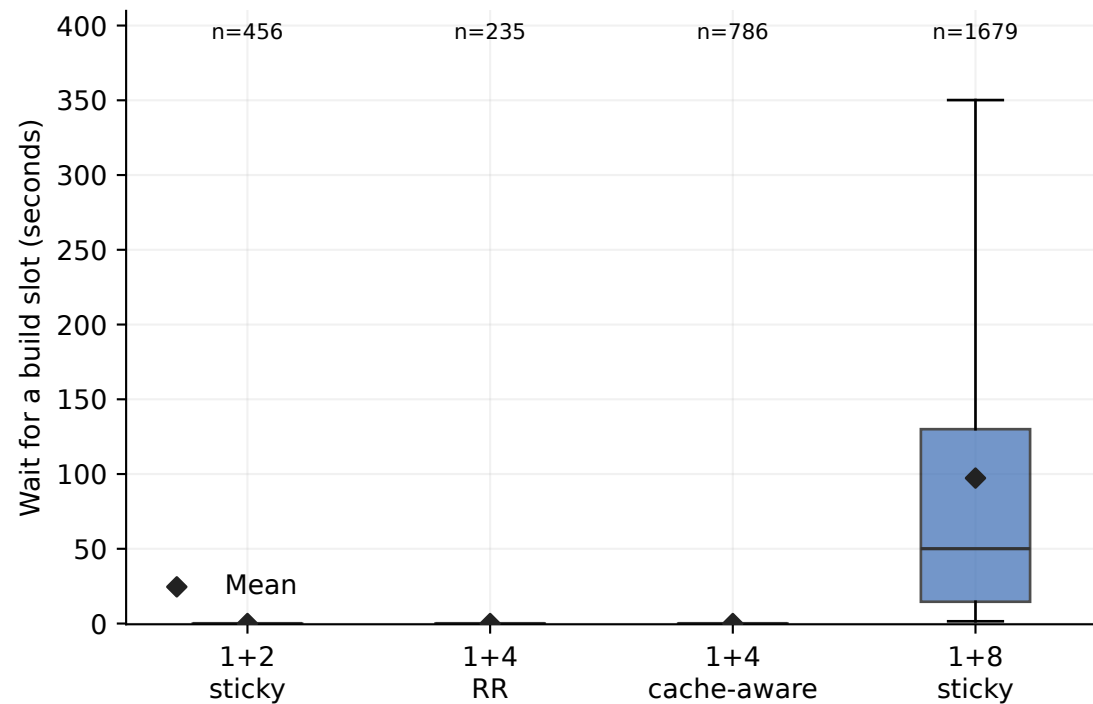


mini-SWE-agent



Top: full-window CPU samples aggregated into 60s means. Bottom: measurement-admission cohort, completed through drain.
Time components are disjoint interval unions; mean bars do not imply identical task composition across runs.

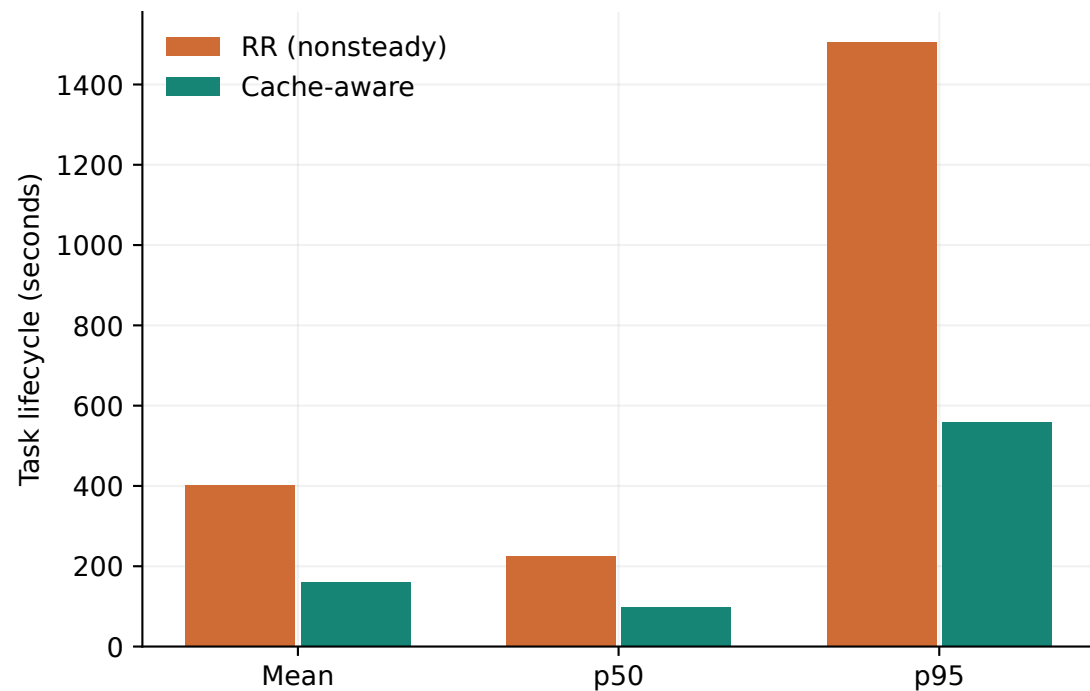
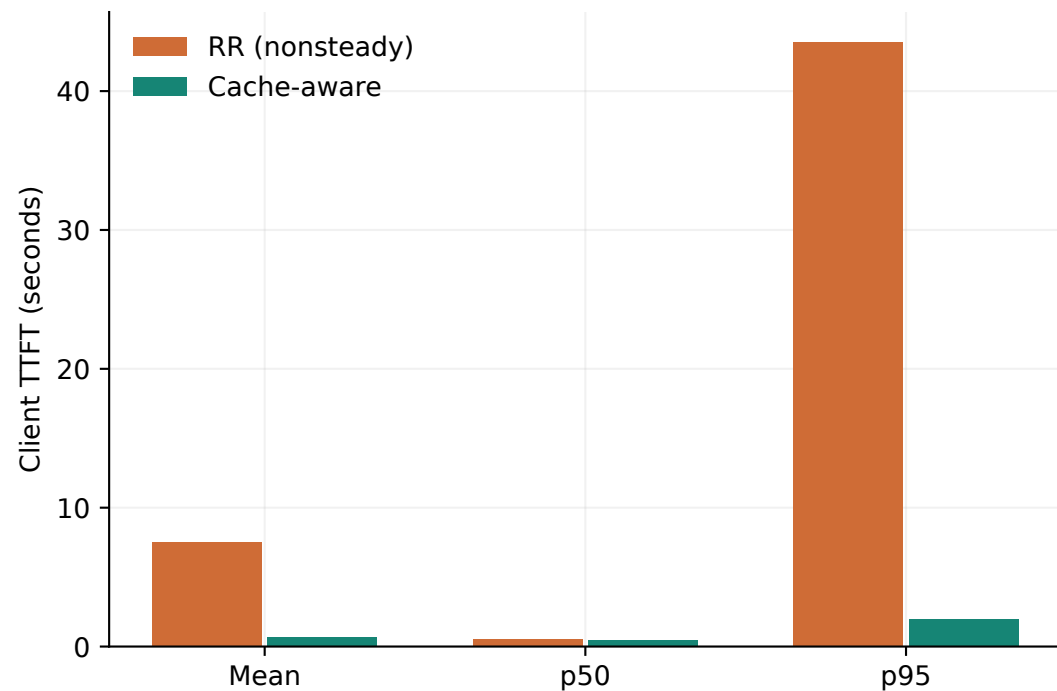
mini-SWE environment setup: slot waiting and build slowdown



All measurement admissions with setup records, including later-cancelled RR tasks; no missing records.

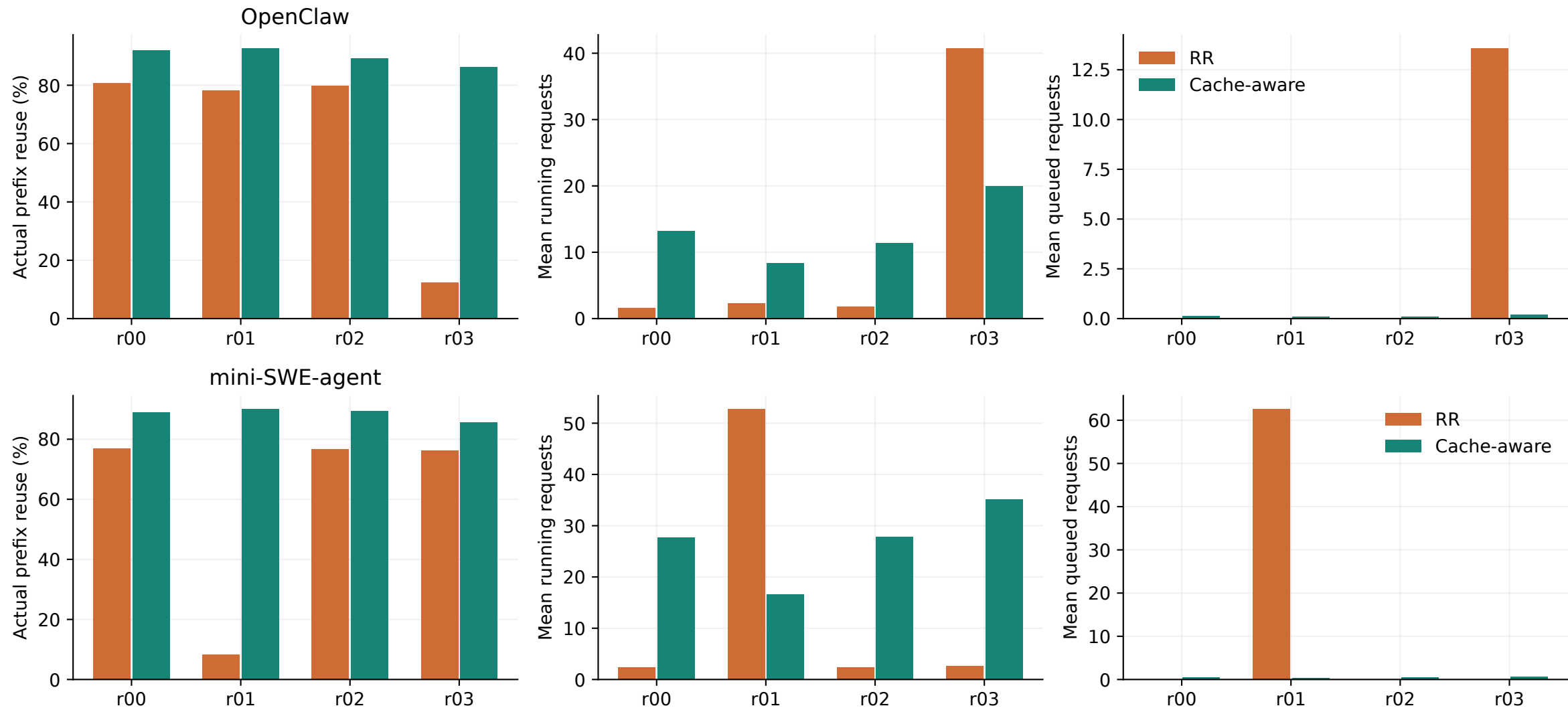
Box: p25-p75, center: median, whiskers: p5-p95; outliers hidden. All four runs allow 4 concurrent builds per frontend.

Completed OpenClaw routing latency cohorts



OpenClaw: full completion through drain; task admissions / LLM starts within the measurement window.
Mini-SWE RR latency omitted: 36 admission-cohort tasks were cancelled after the 30-minute drain limit.

RR hides a hotspot behind similar per-backend output rates



Full-window samples. RR hotspots: OpenClaw r03, mini-SWE r01. Closed-loop task progress can be limited by these replicas. This identifies queue/cache concentration; the initiating cause needs repeats / request-level context and routing diagnostics.